A8. Assembler tilida murakkab arifmetik amallarni bajarish

Amaliy ishning maqsadi

Talabalarda assembler tilida murakkab arifmetik amallarni (ko‘paytirish, bo‘lish, qoldiq topish, ifodalarni hisoblash) bajarish bo‘yicha amaliy ko‘nikmalarni shakllantirish, shuningdek registrlar, flaglar va ALU ning roli bilan yaqindan tanishtirish.

Kerakli jihozlar va vositala

- Kompyuter yoki noutbuk
- (ixtiyoriy) **EMU8086 / DOSBox / NASM** muhiti
- Windows Calculator → *Programmer mode*
- Microsoft Word / Google Docs

Nazariy qisqacha eslatma

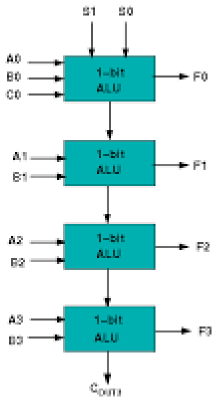
Assembler tilida **oddiy arifmetik** (`ADD`, `SUB`) bilan birga **murakkab arifmetik amallar** ham mavjud. Ular quyidagilar:

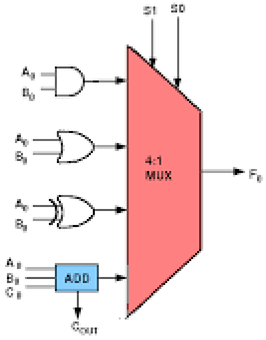
- **ko‘paytirish** (`MUL`, `IMUL`)
- **bo‘lish** (`DIV`, `IDIV`)
- **qoldiqni aniqlash**
- **ko‘p bosqichli ifodalarni hisoblash**

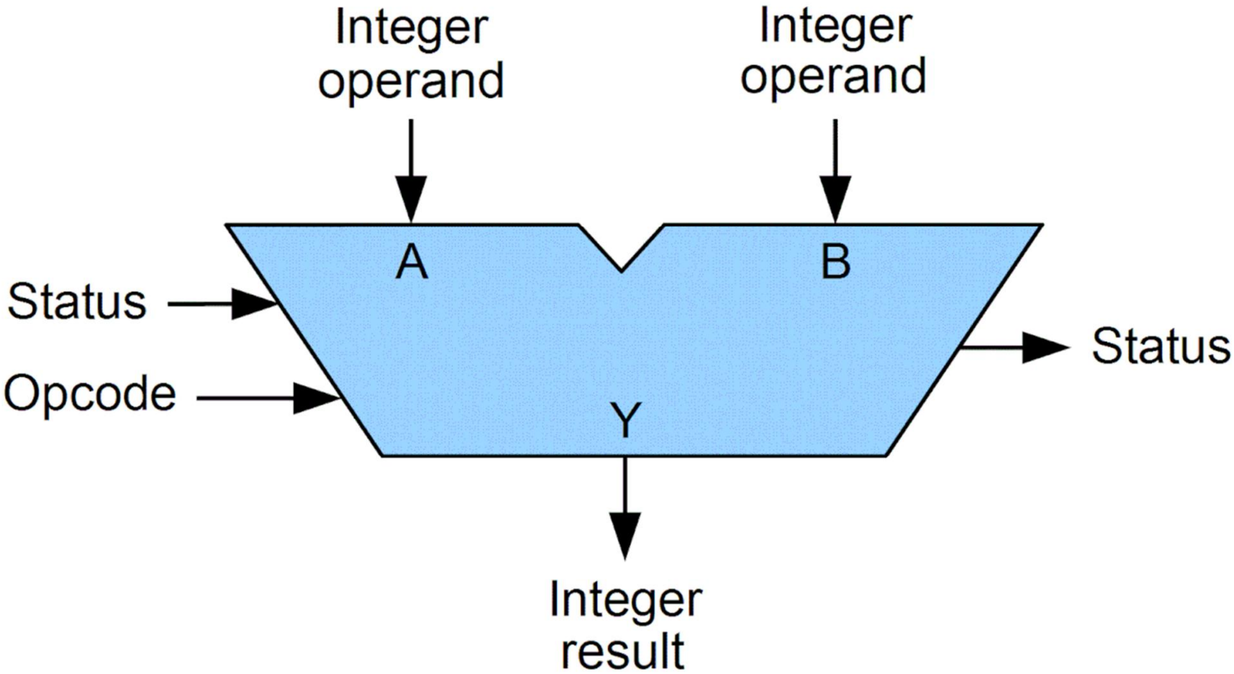
Bu amallarni bajarishda **AX, BX, DX** kabi registrlar va **Carry, Overflow** flaglari muhim rol o‘ynaydi.

1-topshiriq. Murakkab arifmetik amallarni ALU da bajarilishi

Quyidagi rasm asosida murakkab arifmetik amallar qayerda bajarilishini aniqlang.







Topshiriq:

1. Ko‘paytirish va bo‘lish qayerda bajariladi?
2. Qaysi registrlar ishtirok etadi?
3. Nima sababdan DX registri ishlatiladi?

2-topshiriq. Assemblerda ko‘paytirish (MUL, IMUL)

2.1. Belgisiz ko‘paytirish (MUL)

Misol:

```
MOV AX, 6
MOV BX, 4
MUL BX
```

Natija:

- AX = 24
- Agar natija katta bo‘lsa → DX:AX ishlatiladi

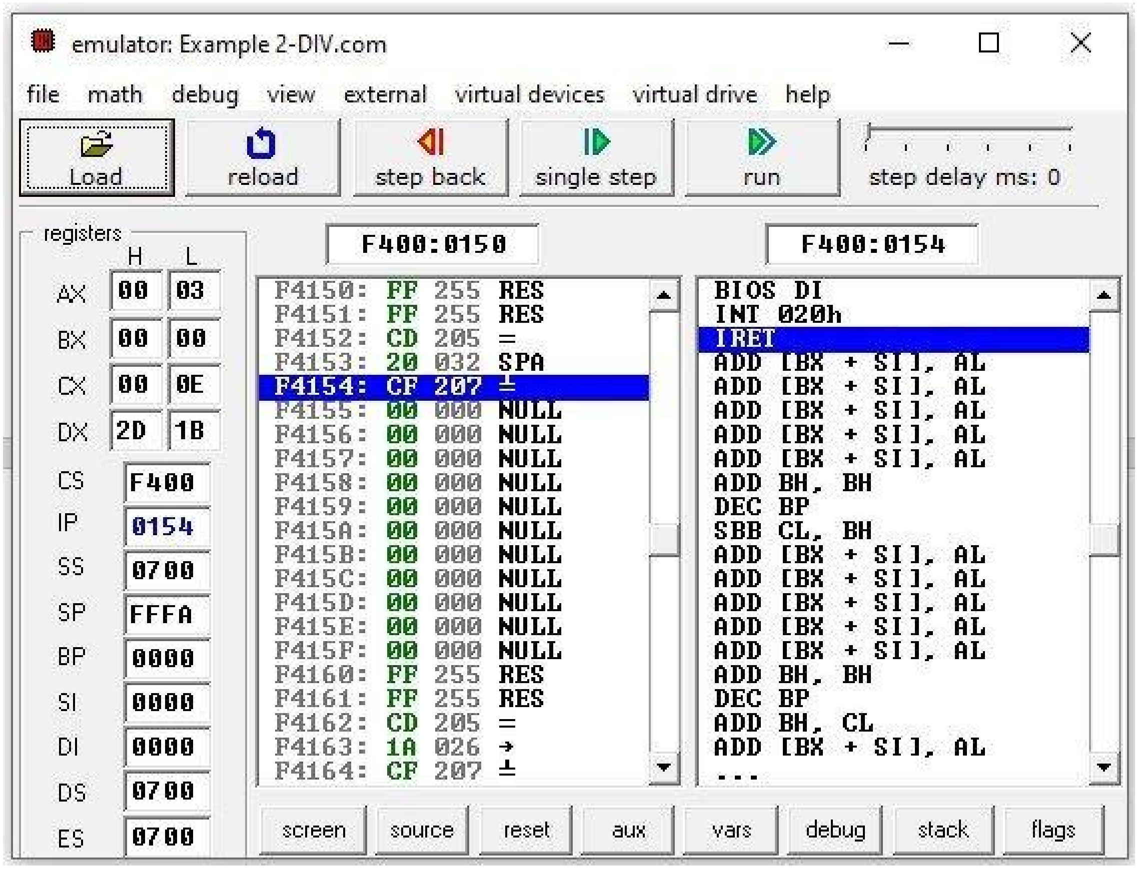
2.2. Belgili ko‘paytirish (IMUL)

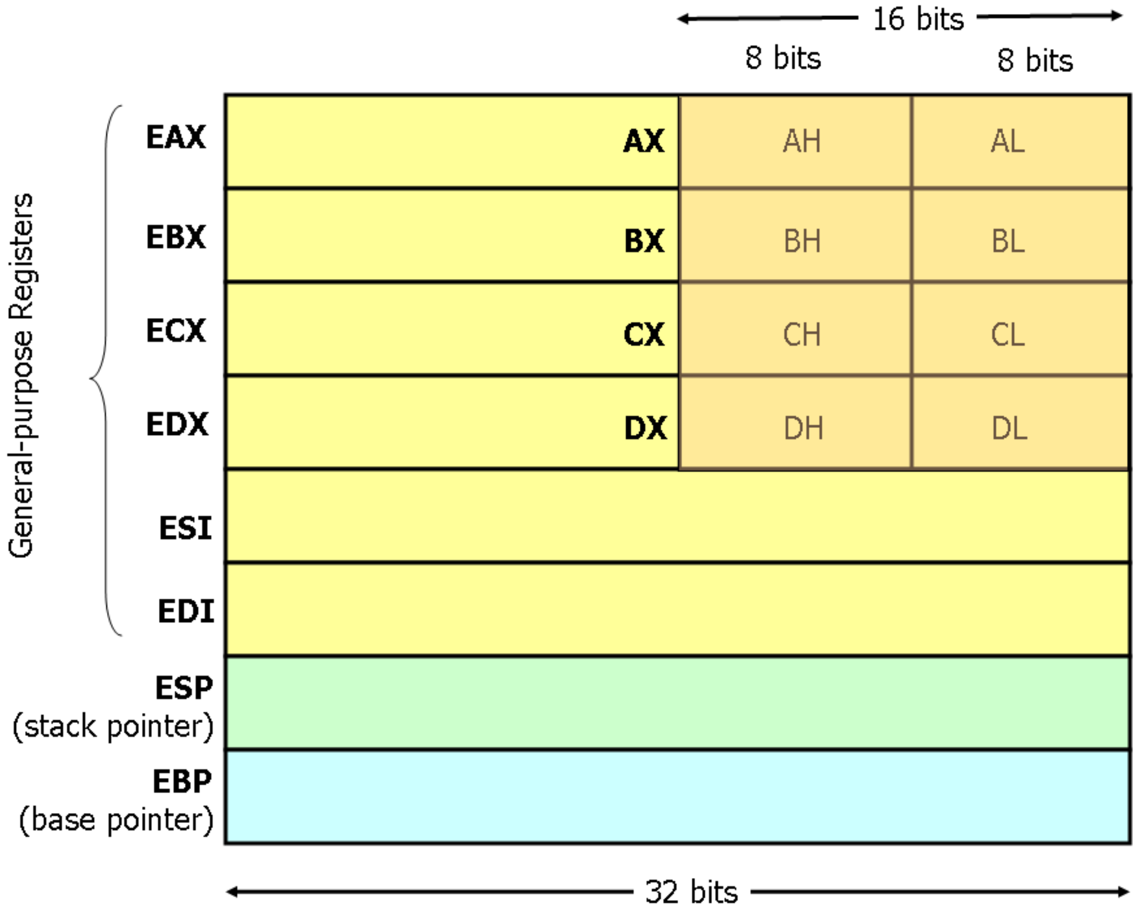
```
MOV AX, -5
MOV BX, 3
IMUL BX
```

Topshiriq: natijani o‘nlikda izohlang.

3-topshiriq. Assemblerda bo‘lish (DIV, IDIV)

Quyidagi rasm orqali bo‘lish mexanizmini tushuning.





3.1. Belgisiz bo‘lish (DIV)

```
MOV AX, 20
MOV BX, 4
DIV BX
```

📌 Natija:

- AX → bo‘linma
- DX → qoldiq

3.2. Belgili bo‘lish (IDIV)

```
MOV AX, -20
MOV BX, 4
IDIV BX
```

📌 **Topshiriq:** qoldiq qayerda saqlanadi?

4-topshiriq. Qoldiqni aniqlash (MOD amali)

Assemblerda alohida `MOD` yo‘q, ammo:

```
DIV BX
; DX da qoldiq
```

📌 **Topshiriq:**
27 MOD 5 ni assembler mantiqida tushuntiring.

5-topshiriq. Murakkab arifmetik ifodani hisoblash

Berilgan ifoda:

(A+B)×C−D(A + B) \times C - D(A+B)×C−D

Assembler mantiqi:

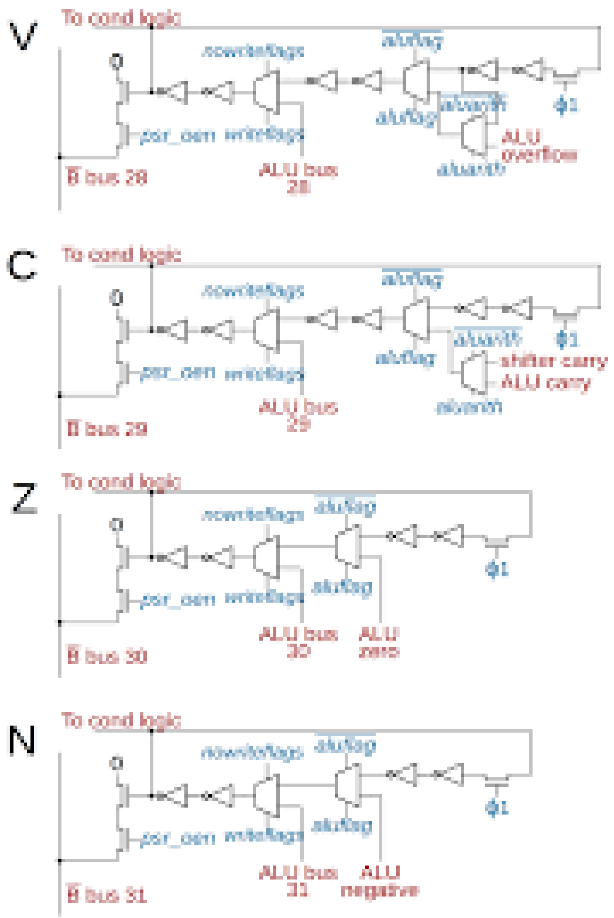
```
MOV AX, A
ADD AX, B
MUL C
SUB AX, D
```

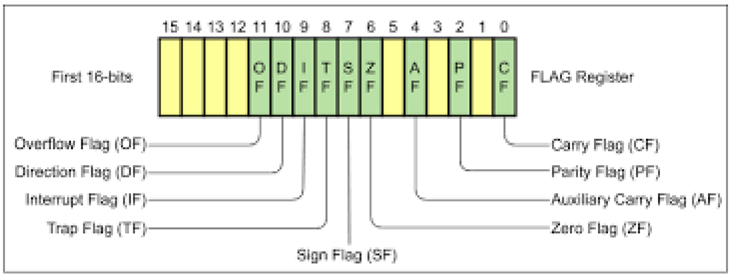
📌 **Topshiriq:**
A=4, B=2, C=3, D=5 bo‘lsa, natijani toping.

6-topshiriq. Flaglar bilan ishlash

Murakkab arifmetik amallar quyidagi flaglarga ta’sir qiladi:

- Carry Flag (CF)
- Overflow Flag (OF)
- Zero Flag (ZF)





Topshiriq:

Qaysi holatda CF yoki OF o‘zgaradi? Misol bilan tushuntiring.

7-topshiriq. Kompyuterda tekshirish (qiziqarli)

1. Calculator → Programmer mode
2. Decimal / Binary rejimlarda
3. Ko‘paytirish va bo‘lishni bajaring

Xulosa yozing:

Assemblerdagi natija bilan mos keldimi?

8-topshiriq. Tahliliy savol (eng muhim)

Quyidagi savolga yozma javob bering:

Nima sababdan assemblerda murakkab arifmetik amallar maxsus buyruqlar orqali bajariladi?

Kamida **3 asos** yozilsin.

Hisobot tarkibi (talaba topshiradi)

1. Amaliy ish nomi (A8)
2. Maqsad
3. MUL/DIV tushunchasi
4. Amaliy kodlar
5. Ifoda hisoblash
6. Flaglar tahlili
7. Xulosa

Xulosa (namuna)

Ushbu amaliy ish davomida assembler tilida murakkab arifmetik amallarni bajarish o‘rganildi. Ko‘paytirish, bo‘lish va qoldiqni aniqlash jarayonlari registrlar va flaglar orqali tahlil qilindi. Natijada assembler tilining past darajadagi hisoblash imkoniyatlari aniq tushunildi.

Nazorat savollari

1. MUL va IMUL farqi nimada?
2. DIV buyrug‘idan keyin qoldiq qayerda saqlanadi?
3. Nima sababdan DX:AX jufti ishlatiladi?
4. Murakkab ifodalar qanday ketma-ketlikda hisoblanadi?
5. Flaglar arifmetik amallarga qanday ta’sir qiladi?